Cambridge Permanence and Durability Workshop

January 31–February 1, 2024

Workshop Resources and Summary

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Permanence Resources

Historical Context

Carbon markets and emissions trading systems seek to leverage markets to incentivize and mobilize finance for cost-effective, standardized and verified greenhouse gas emission reductions and carbon removals. Carbon markets broadly refer to “cap and trade” and other emissions trading systems, as well as voluntary and compliance “offset” markets that generate tradeable certificates representing one tonne of CO2e avoided, reduced, or removed from the atmosphere.

Demand for these certificates is driven by either 1) compliance obligations on companies or governments as a result of regulation or 2) voluntary actors seeking to deliver on climate action goals or targets.

Conventions and terminology have changed over the years but generally speaking, a “carbon credit” represents one metric tonne of CO2e avoided, reduced or removed emissions from the atmosphere. Most carbon credits are used as a carbon “offset”, e.g. to compensate for a tonne of emissions elsewhere, either under mandatory or voluntary systems. If a carbon credit is claimed as an offset, there’s a general expectation in the market that purchasing and retiring a carbon offset delivers an equivalent or commensurate amount of atmospheric benefit/climate change mitigation benefit to the tonne of emissions you are compensating for.

The concept of “permanence” has long been considered a key tenant of a high-quality carbon credit. “Permanence” is the concept that carbon removed or reduced through an offset project must have a high degree of certainty that it will last for a sufficient period of time to account for/compensate for the harm of the impact of carbon dioxide in the atmosphere it is intended to address.

However, to date, carbon credits considered “permanent” have been issued from a range of project types resulting in carbon removed or reduced for different periods of time—ranging from a year to centuries. This lack of consistent accounting for differences in storage or reduction duration across project types, and the lack of consensus across the policy making and scientific fields about the appropriate period of time required to store carbon in order to fully mitigate the negative impacts of the compensated tonne, has led to concerns over the validity of some claims based on carbon credits and uncertainty around how to value the climate impact from different projects of different lengths. Despite broad recognition of the atmospheric value and benefit of temporary removal and storage of carbon, there is not yet consensus on how to value these types of projects relative to projects that store carbon for much longer terms.

Recently, “durability” has gained traction as an alternative to “permanence.” This represents a recognition that shorter term solutions have value in climate mitigation and that a range of approaches are available to value and conduct accounting for the duration of impact (i.e., minimizing the risk that the impact is reversed or not achieved for the length of time we initially believe it to be) rather than a minimum time period that all projects must adhere to. Others in the market insist that only solutions that last as long as carbon in the atmosphere (i.e., between 300 and 1,000 years) have merit.

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Current approaches to permanence across voluntary and compliance standard setting bodies vary widely with respect to classification of risk across mitigation project types, how removals and reductions are accounted for, risk mitigation and management assessments and systems, length of monitoring, reporting and verification periods, etc. This is a key policy issue but has historically been driven by individual standard bodies and mechanisms and addressed piecemeal across the market. Notably California has extended the disclosure requirements to durability and risk mitigation and specified these be "for any project that the business entity knows or should know that such reductions or removal enhancements are less than the atmospheric lifetime of carbon dioxide emissions."

Incumbent ratings entities, data providers, insurers and price assessment agencies evaluate duration without a common framework/s to compare projects of different durations. Rather, proxies are used for data such as credit type and vintage.

The European emission trading scheme appears to favor a conservative approach to equivalence for VCCs and exclusion of any carbon credits, even if permanent. A recent CO2RE report called for the focus on monitoring reporting and verification be applied to all removals 'regardless of whether of differing permanence and policy/market design. Although the UNFCCC Supervisory Body for Article 6.4 decided in 2023 to 'focus on measures that address reversals on a tonne-for-tonne basis and drop tonne-year carbon accounting method (tonne-year carbon accounting method discussed below) from near-term considerations' work is ongoing and Article 6.4 agreements pushed out for another year.

Permanence and Durability Approaches

Approaches to address permanence and durability risks in the voluntary carbon market range from wholesale changes to the way carbon storage is accounted for to more targeted risk mitigation measures that can be aligned to most existing standards bodies today. Below is an overview of sample approaches to address permanence and durability in the voluntary markets that may be discussed more in-depth at the workshop.

The descriptions below do not include commentary on the market demand for such approaches but focus on how they address permanence or durability risk in the supply of carbon credits. This is not to say that demand is not important. On the contrary, market demand is a vital consideration when determining the feasibility of implementing any given approach along with other considerations (see Permanence and Durability Parameters). Furthermore, how the "demand" side of the market, e.g. carbon credit purchasers and investors, sees requirements imposed by standard setting bodies and regulators has a significant impact on whether a given approach to permanence will be successful over time. Feasibility of application and acceptability by "demand" side actors should be taken into consideration.

IC-VCM, despite its focus on the supply side, regularly coordinates with other ecosystem actors more focused on the demand side of the market, including the Voluntary Carbon Markets Initiative, We Mean Business Coalition, Climate Disclosure

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Project, Science Based Targets Initiative, and Greenhouse Gas Protocol (see recent, relevant examples from COP28 here and here).

In addition, the descriptions below do not explicitly identify assignment of liability for reversal risk. For any approach, building a system that will attract investment from traditional capital markets will require clarity on which market actor bears liability in the event of a default as well as available recourse. Currently, there is not a standard approach to assigning responsibility for compensating reversals. Market actors have long advocated for seller liability and the majority of more prominent standards and protocols in the voluntary carbon market space rely on seller liability for any reversal (discussed in more detail below).

For additional reading regarding each approach, please reference the Permanence and Durability Index in the appendix.

Supply Side Targeted Risk Mitigation Mechanisms

These approaches to address permanence and durability risks are focused specifically on the supply side of the market and involve little change in how buyers would account for their emissions.

Minimum Project Duration with Monitoring, Reporting, and Verification (MRV), Compensation, and Liability Assignment

It is common practice in the carbon crediting markets to require a monitoring, reporting, and verification (MRV) and compensation period during which project developers must regularly monitor for any reversals and implement mitigation or compensation measures if the emissions underlying a previously issued credit are reversed.

Each standard takes a different approach to determining the length of their MRV and compensation period, typically based on a combination of project type risk and individual project risk assessments. For example, Climate Action Reserve's U.S. Forest Protocol specifies that forest projects must be monitored and verified "for a period of 100 years following the issuance of any [climate reserve tonne] for GHG reductions or removals achieved by the project." Verra requires its agriculture, forestry, and other land use projects to assess permanence risk over 100 years; however, it requires a minimum project duration of only 40 years. To account for this difference, Verra is developing a long-term monitoring system for projects certified under its methodologies that would continue to monitor reversal risk after the crediting period or MRV and compensation period expires.

MRV and compensation periods for engineered removals vary depending on project type. For example, geologic storage protocols often point to the U.S. Environmental Protection Agency's guidance around class VI wells for carbon dioxide sequestration, which requires a minimum 50-year monitoring period after the injection well is closed. That being said, while government regulations may require an MRV period, protocols may not require compensation in the event of a reversal for engineered solutions—even during this MRV period.

In addressing permanence and durability risk in its Assessment Framework, IC-VCM provided guidance around MRV and compensation periods by activity type. For activities that are considered to have a "material risk" of reversal (i.e., conservation and

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avoided conversion, agriculture soil carbon sequestration, forestry sequestration, and wetland and marine ecosystem restoration/management), IC-VCM requires a minimum 40-year monitoring and compensation period. For projects that are considered to have a lower risk of reversal (i.e., displacement of non-renewable biomass, biochar, carbon capture with geological storage or mineralization, enhanced weathering, carbon dioxide in concrete) IC-VCM requires standards to conduct a general risk assessment and implement appropriate measures to mitigate any material risks. For Jurisdictional REDD+ projects, IC-VCM requires a minimum 40-year monitoring and compensation period.

In the next iteration of its Assessment Framework, IC-VCM noted that it would specifically consider two changes to its guidance:

- Extending the monitoring and compensation period (e.g., 100 years)
- Shifting the monitoring and compensation oversight to the carbon-crediting program or relevant jurisdiction

Buffer Pools

Buffer pools are the most commonly used tool to address permanence and durability risks in the market today and were specifically addressed by IC-VCM in its Assessment Framework. Buffer pools are a mechanism whereby a portion of a project's overall credits are reserved to cover unplanned reversals and ensure that any specific entity (whether a project or a credit issuer or otherwise) has reduced or removed more carbon dioxide from the atmosphere than it has released. Buffer pools are used by the largest registries today (e.g., Verra, Gold Standard, American Climate Registry, Climate Action Reserve, etc.) and are typically managed by the registries themselves; however, each standard takes a slightly different approach to buffer pool management.

In its Assessment Framework, IC-VCM provided guidance for when buffer pools are required and how they should be structured. For projects that are considered to have a material risk of reversal (i.e., conservation and avoided conversion, agriculture soil carbon sequestration, forestry sequestration, and wetland and marine ecosystem restoration/management), IC-VCM requires a minimum of 20% of all credits issued by the standard to be placed in a pooled buffer reserve. For projects that are considered to have a lower risk of reversal (i.e., displacement of non-renewable biomass, biochar, carbon capture with geological storage or mineralization, enhanced weathering, carbon dioxide in concrete) IC-VCM requires standards to conduct a general risk assessment and implement appropriate measures to mitigate any material risks. For Jurisdictional REDD+ projects, IC-VCM requires a pooled buffer reserve but leaves discretion on the adequate size, noting that it should be "proportional to reversal risk and adequate to compensate for potential reversals."

In the next iteration of its Assessment Framework, IC-VCM noted that it would consider requiring credit issuers to have provisions in place to ensure the continued operation of pooled buffer reserves until the latest expiration of any given project (e.g., in case the credit issuer ceases to exist or can no longer operate the pooled buffer reserve).

Insurance

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While some consider buffer pools a form of insurance, other third-party insurance offerings have begun to appear. Insurance policies targeted at the voluntary carbon market are being offered by large more traditional insurers and startups alike; however, policies that cover risk related to permanence and durability remain few and far between. Several insurers have chosen to enter the market focused on mitigating other risks associated with carbon credit transactions, including political risk and delivery risk, among others.

Insurance policies that address permanence and durability could be targeted at project developers, carbon credit buyers, and credit issuers. For project developers, an insurance policy could cover an unavoidable reversal after credits have been issued (e.g., an afforestation project damaged by wildfire or pest infestation or a carbon dioxide injection site with a caprock damaged by unrelated seismic activity). For carbon credit buyers, an insurance policy could cover a portfolio of credits purchased and be priced according to the activities associated with the credits purchased. For credit issuers, an insurance policy could cover the pooled buffer reserve, providing issuers with the ability to compensate those who purchased the credits.

An open question among insurers related to permanence and durability is that of fungibility. Given that buyers are often purchasing a carbon credit to offset an emission, monetary compensation may not be the preferred remedy. Instead, insurers are exploring the possibility of paying out policies in carbon credits, but this introduces a new set of issues to be addressed—should the replacement credit be from the same activity? Should it be from the same issuer? Should the replacement credits be from a project in the same geography? What are the ecological impacts associated with the different options? These questions and more are still being debated, with insurers developing their own approaches to fungibility.

Discounting

Discounting is another approach that is used to account for permanence and durability risk as well as other risk factors associated with a carbon credit. There are both ex-post and ex-ante examples of discounting being applied to carbon credits. It is common practice for crediting methodologies and standards to require a variety of discounts to be applied prior to credit issuance to address a range of issues, including uncertainty and to ensure conservativeness. In some instances buyers and developers have applied an ex post credit discount, where a credit that is issued as equal to a metric tonne of carbon dioxide removed from the atmosphere is multiplied by a percentage somewhere between 0% and 100% to become worth less than one metric tonne of carbon dioxide removed according to its individual risk profile as determined by the buyer or broker. This method is extended to the project level, where the overall volume of issued credits is discounted to account for permanence, durability risk, and other associated risks.

Risk-adjusted discounting is commonly used in financial markets and has its origins in the capital asset pricing model first published in 1964 and is still frequently used to value future cash flows. While risk-adjusted discounting has been a concept discussed in the carbon markets since shortly after the adoption of the Kyoto Protocol, it has more recently been a feature of how carbon credit ratings agencies assign value to a carbon

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credit. For example, Sylvera has encouraged its customers to use its ratings to calculate a discount rate, while BeZero has introduced a framework that applies a discount factor to its ratings. Risk-adjusted discounting has also been used on the demand side by carbon credit intermediaries like Rubicon Carbon as a tool to manage the over-crediting risk of projects across a portfolio.

Finally discounting is also used to calculate and assign value to emissions avoided or reduced in monetary terms (typically structured as a "social cost of carbon" calculation) to enable assignment of value to various durations of carbon storage in economic terms and to derive a net present value of the avoided damages of emissions in the future.

Other activity and methodology-specific mechanisms

There are several activity-specific mechanisms that could be implemented at the project level to aid in mitigating permanence or durability risk. Each of these mechanisms will be specific to the particular activity type and are typically addressed through project specific methodology requirements.

For example, in its geologically stored carbon methodology, Puro.Earth requires that carbon dioxide be sequestered in reservoirs or geological formations that meet U.S. Environmental Protection Agency class I, class II, class VI, or EU regulations. The Gold Standard, requires that, with limited exceptions, calcium carbonate from indirect mineral carbonation must be used as filler in construction materials only to avoid accidental release through municipal solid waste incineration. Verra recently concluded a public comment period for its proposed ABACUS label for afforestation, reforestation, and revegetation activities, outlining several project-level measures to enhance permanence and durability. The project working group proposed mandating that projects be "ecologically appropriate" for the geographic area, disallowing projects with more than one contiguous hectare of monoculture, and requiring at the project validation phase a proposed approach to ensure carbon stock stability after the crediting period (e.g., legal protection, ongoing capacity building, alignment of financial incentives, etc.).

Different Approaches to Account for Carbon Emissions

These approaches differ from those outlined above in that they could require a change in how purchasers approach carbon credit use and accounting to address permanence and durability risk in carbon credit projects.

Tonne-Year Accounting

Tonne-year accounting refers to a family of approaches that seek to introduce time into how carbon emissions are accounted for, providing market participants with a common unit (a tonne-year) and method to compare the impact and durability of different credited emission mitigation activities with the impact and duration of emissions from elsewhere in the economy. In the voluntary carbon markets, the application most associated with tonne-year accounting is known as vertical stacking.

Vertical stacking refers to "buying multiple short-termne-year accounting aims to provide a means to account for and compare the impact of temporary carbon storage and projects of different durations. " credits upfront to "offset one tonne of carbon dioxide emitted today." For example, to offset one agetonne emitted in 2020, an organization would purchase multiple credits with a 20-year duration to attempt to approximate the

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impact of the carbon emitted in 2020. One of the more significant challenges with vertical stacking is determining how high the stack should be—e.g., is it two 20-year credits? Three? More? Two prominent approaches for calculating the value of temporary storage in a vertical stack, Moura-Costa and Lashof, were first introduced in the 1990s, while more recent research has sought to reframe the conversation around the social value of an offset.

While the IC-VCM did not explicitly address tonne-year accounting in its Assessment Framework, it noted during its feedback summary that tonne-year accounting approaches without any minimum time commitment would not be granted Core Carbon Principles status. The expert panel acknowledged that tonne-year accounting in some form "could work as an incentive mechanism or policy tool for maintaining carbon storage;" however, it rejected the "flawed premise that storing carbon for arbitrarily short periods of time" could be deemed equal to a permanent reduction in emissions.

Temporary Crediting

Often linked with tonne-year accounting, but arguably a distinct concept is the idea of temporary crediting. Temporary crediting frequently relies on some of the same scholarship (i.e., Moura-Costa, Lashof, Groom and Venmans) in seeking to arrive at a value for short-term carbon storage and then allows for a credit "expiration" after a given time interval. What happens upon expiration is still up for debate—carbon credit buyers could then replace the credit with another one, a public entity could manage the replacement of expiring credits, etc.—the key element is just that the credit itself is time-bound.

Temporary crediting could have positive atmospheric and social benefits. In terms of atmospheric benefits, it could give "ecosystems more time to adapt to climate change" and also provide society with more time to "develop and deploy alternative mitigation actions." From a social perspective, temporary crediting could direct capital to indigenous peoples and local communities who are often intimately involved in projects that issue temporary credits (e.g., reforestation and ecosystem restoration projects).

Permanent Additional Carbon Tonne

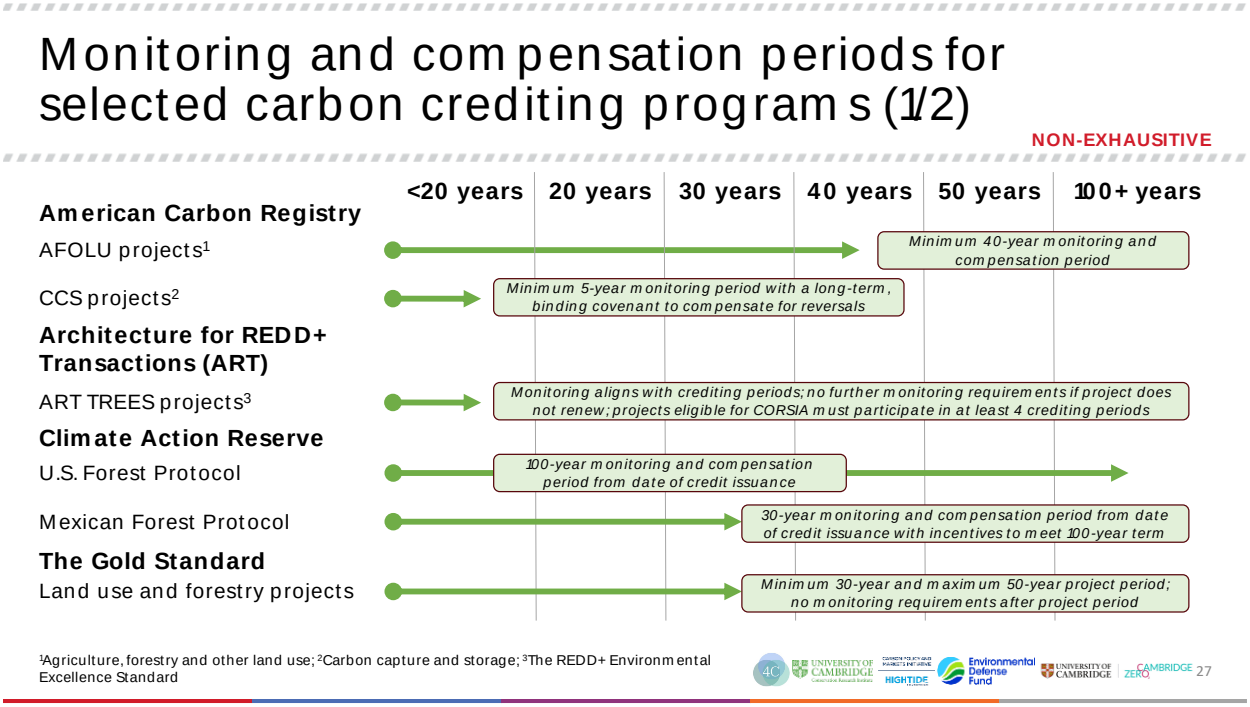
The Permanent Additional Carbon Tonne (PACT) is a framework for quantifying the long-term economic value of nature-based solutions to compare directly with carbon capture and storage or geosequestration. It was developed by a team of Cambridge University researchers and first published in October 2023. The framework includes six elements: additionality, permanence, leakage, biodiversity, livelihoods, and justice.

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Current Practice Among Carbon Crediting Programs

Current approaches to permanence and durability vary among carbon crediting programs and activity type. With the release of the IC-VCM's Core Carbon Principles last year, some carbon crediting programs have updated their methodologies to meet IC-VCM standards, while others were already meeting or exceeding IC-VCM guidance.

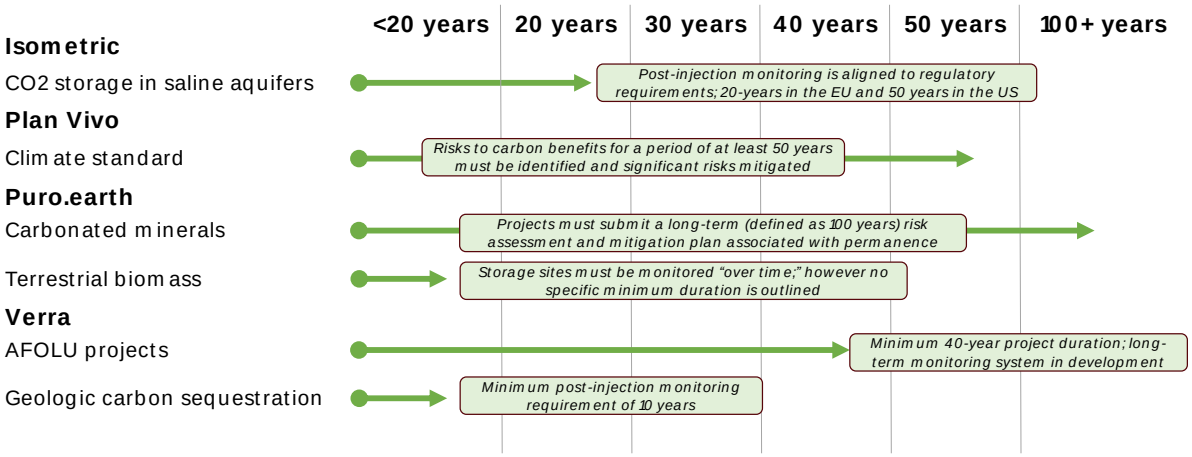
The research included below is non-exhaustive and does not represent all carbon crediting programs or all activity types for each carbon crediting program. The research is meant to highlight how a few carbon crediting programs approach issues of permanence and durability in different activity types and is not representative of any formal assessment by the IC-VCM. Additional relevant detail related to each carbon crediting program is included after the summary graphics.



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Monitoring and compensation periods for selected carbon crediting programs (2/2)

NON-EXHAUSTIVE



Other reversal risk mitigation measures by carbon crediting program (1/2)

NON-EXHAUSTIVE

	Program uses a pooled buffer reserve to compensate for reversal	Fraction of credits in buffer is standard % of total issued carbon credits	Fraction of credits in buffer specific to individual project risk	Program requires or incentivizes mitigation of reversal risk	Easily findable, publicly available information on buffer contents	Provisions to operate reserve if Program ceases to exist
American Carbon Registry						
AFOLU projects	✓		✓	✓	✓	
CCS projects	✓	✓		✓	✓	
Architecture for REDD+ Transactions (ART)						
ART TREES projects	✓		✓	✓	✓	
Climate Action Reserve						
U.S. Forest Protocol	✓		✓	✓		✓
Mexican Forest Protocol	✓		✓	✓		✓
The Gold Standard						
Land use and forestry projects	✓	✓		✓		

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Other reversal risk mitigation measures by carbon crediting program (2/2)

NON-EXHAUSTIVE

	Program uses a pooled buffer reserve to compensate for reversal	Fraction of credits in buffer is standard % of total issued carbon credits	Fraction of credits in buffer specific to individual project risk	Program requires or incentivizes mitigation of reversal risk	Easily findable, publicly available information on buffer contents	Provisions to operate reserve if Program ceases to exist
Isometric						
Direct air capture + injection	✓	✓		✓		
Plan Vivo						
Climate Standard	✓	✓		✓	✓	
Puro.earth						
Carbonated minerals				✓		
Terrestrial biomass				✓		
Verra						
AFOLU projects	✓	✓		✓	✓	
Geologic carbon sequestration	✓	✓		✓	✓	

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American Carbon Registry

The American Carbon Registry's reversal risk mitigation requirements are specific to the activity type, with AFOLU projects mirroring IC-VCM's 40-year monitoring and compensation period requirement. Across both AFOLU and CCS projects, ACR requires project developers to contribute to a pooled buffer reserve that can be used as compensation in the event of a reversal risk. The contribution from AFOLU projects is determined based on a risk assessment, with lower risk factors leading to a lower required contribution. For CCS projects, the requirement is a flat 10% with a required binding risk mitigation covenant. No information regarding contingency plans for the continued operation of the pooled buffer reserve was found.

Architecture for REDD+ Transactions (ART)

The ART TREES program requires a reversal risk assessment to determine a buffer contribution that is tailored to the individual jurisdiction. Reversal risk is evaluated every five years; however, should a jurisdiction withdraw from the program, no further monitoring related to already issued credits is required. No information regarding contingency plans for the continued operation of the pooled buffer reserve was found.

Climate Action Reserve

Both of Climate Action Reserve's protocols require a reversal risk assessment to determine the required buffer contribution, with the contribution lower if more risk mitigation measures are included in the project design. Climate Action Reserve indicates that its bylaws require the continued operation of the pooled buffer reserve should the organization cease to exist, but the exact pathway is left undetermined and dependent on future market conditions. The program may publish information regarding

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its pooled buffer reserve contents in the login-only section of its website; however, no information was found in investigating the public portions of its website.

The Gold Standard

Unlike other carbon crediting programs, Gold Standard requires a flat 20% contribution to its pooled buffer reserve by all land-use projects, with other built-in requirements that seek to minimize reversal risk. Gold Standard was also one of the few standards to set a maximum project duration period for land-use projects. No information regarding contingency plans for the continued operation of the pooled buffer reserve was found.

Isometric

Isometric requires a small, 2% contribution to a pooled buffer reserve as a reversal risk mitigation measure, even though the program focuses on methodologies that achieve "effective" permanence. The program requires other risk mitigation measures aligned to state or national regulations for its saline aquifer storage module. No information regarding contingency plans for the continued operation of the pooled buffer reserve or information on the pooled buffer reserve's contents was found.

Plan Vivo

Plan Vivo requires a flat 20% of all issued credits to be transferred to its risk buffer and requires mitigation of significant reversal risks for at least 50 years. Plan Vivo's standard also incorporates several other requirements aimed at further mitigating reversal risk. Plan Vivo has published the number of credits in its pooled buffer reserve for previous protocols; however, it has not updated its website with information for the most recent versions. No information regarding contingency plans for the continued operation of the pooled buffer reserve was found.

Puro.Earth

Puro.Earth was the only carbon crediting program researched that does not have a pooled buffer reserve to manage reversal risk. Instead, it relies on a combination of monitoring requirements, documentation requirements (e.g., land title in the case of terrestrial biomass storage) and pre-issuance deductions to manage reversal risk.

Verra

Verra recently updated some of its guidance to mirror the permanence requirements contained in the IC-VCM's Assessment Framework. Verra is the only standard that is building a long-term monitoring capability that will eventually be deployed to monitor for reversal risk after the project period concludes. Verra requires that every program conduct a risk assessment to determine a buffer pool contribution, with a lower contribution tied to strong risk mitigation measures. Of all carbon crediting programs investigated, Verra provided the most comprehensive account of its pooled buffer reserve contents, with data down to the project level publicly available. No information regarding contingency plans for the continued operation of the pooled buffer reserve was found.

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Permanence and Durability Parameters

Workshop attendees will discuss and consider approaches to permanence and durability in practice today and novel proposed approaches such as those described above. These Parameters aim to describe the attributes of the ideal permanence and durability system for the carbon market. and assist attendees in the systems design exercise.

As workshop attendees discuss different approaches and complete their systems design exercise, they are encouraged to reference these parameters and provide a subjective evaluation of any approach or recommendation.

Category	Parameter	Evalutation
Science	Approach provides quantifiable certainty of atmospheric impact and has credible scientific literature supporting it	Uncertain Certain 1 2 3 4 5
Policy	Is “standardizable” across geographies, legal systems and project types	Uncertain Certain 1 2 3 4 5
Enforceability	Enables binding legal contracts and clear assignment of liability	Uncertain Certain 1 2 3 4 5
Accounting	Approach enables projects of different durations to be compared against one another	Uncertain Certain 1 2 3 4 5
Feasibility	Approach could be implemented and adopted at scale (i.e. aligns with how financial markets, legal contracts, and accounting frameworks function)	Uncertain Certain 1 2 3 4 5
Risk Mitigation	Approach incorporates or has identified risk mitigation strategy(ies) to ensure impact	Uncertain Certain 1 2 3 4 5
IPs & LCs	Approach lowers barriers to IPs & LCs participation and adoption	Uncertain Certain 1 2 3 4 5

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Supply	Approach increases the supply of carbon reductions and removals that lead to positive atmospheric impact of reasonable certainty	Uncertain Certain 1 2 3 4 5
Demand	Approach increases the demand of carbon reductions and removals that lead to positive atmospheric impact of reasonable certainty	Uncertain Certain 1 2 3 4 5
Bankability	Approach encourages private-sector investment in a variety of structures	Uncertain Certain 1 2 3 4 5
vali	Approach aligns incentives among different stakeholders	Uncertain Certain 1 2 3 4 5

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Permanence and Durability Source Index

Supply Side Targeted Risk Mitigation Mechanisms

Buffer Pools

Badgley, Grayson, Freya Chay, Oriana S. Chegwidden, Joseph J. Hamman, Jeremy Freeman, and Danny Cullenward. "California's forest carbon offsets buffer pool is severely undercapitalized." *Frontiers in Forests and Global Change* 5 (August 2022). https://www.frontiersin.org/articles/10.3389/ffgc.2022.930426/full.

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IC-VCM Glossary

The glossary below was developed by the IC-VCM and released in tandem with their Core Carbon Principles and Assessment Framework.[1]

Baseline Scenario
A description of the situation and the outcome that is predicted or assumed to occur in the absence of the incentives created by the carbon credits and their associated mitigation activities, while holding all other factors constant.

Benefit Distribution Mechanism(s)
The system(s) or channel(s) through which Monetary and/or Non-Monetary Benefits are distributed.

Benefit Sharing
The sharing of Monetary and Non-Monetary Benefits with mitigation activity's Partners.

Benefit Sharing Arrangement(s)
The arrangement(s) that describe Partners, Monetary and Non-Monetary Benefits, and the Benefit Distribution Mechanism(s). Benefit Sharing Arrangements describe the processes for the distribution of Monetary and Non-Monetary Benefits to Beneficiaries, including the types and proportions of benefits to be shared and the Mechanism by which such benefits will be distributed. Benefit Sharing Arrangements are included in a part of mitigation activity documentation and form the basis of the development of the Benefit Sharing Plan.

Carbon Credit
A tradable financial instrument that is issued by a carbon-crediting program. A carbon credit represents a greenhouse gas emission reduction to, or removal from, the atmosphere equivalent to one metric tonne of carbon dioxide equivalent, calculated as the difference in emissions from a baseline scenario to a project scenario. Carbon credits are uniquely serialized, issued, tracked and retired or administratively cancelled by means of an electronic registry operated by an administrative body, such as a carbon-crediting program.

Carbon-Crediting Program
A standard setting program that registers mitigation activities and issues carbon credits.

Conservative Approaches
A conservative approach errs on the side of caution to ensure emission reductions or removals are unlikely to be overestimated.

Crediting Period
The period in which verified emission reductions or removals attributable to a mitigation activity can result in the issuance of carbon credits.

Double Claiming
A type of double counting in which the same emission reduction or removal is claimed

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[1] "Terms & Definitions." Integrity Council for the Voluntary Carbon Markets. Last modified July 2022. https://icvcm.org/wp-content/uploads/2022/07/ICVCM-Public-Consultation-FINAL-Part-5.pdf.

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by two different entities towards achieving mitigation targets or goals: once by a country, jurisdiction or other entity that reports lower emissions or higher removals for the purpose of demonstrating achievement of a mitigation target or goal, and once by the entity retiring the carbon credit for the purpose of making an emission offsetting claim.

Double Counting
A situation in which a single emission reduction or removal is counted more than once towards achieving mitigation targets or goals. Double counting can occur through double issuance, double use, and double claiming.

Double Issuance
A type of double counting in which more than one carbon credit is issued for the same emission reduction or removal. Double issuance leads to double counting if more than one of these carbon credits is claimed towards achieving mitigation targets or goals.

Double Use
A type of double counting in which a single carbon credit is claimed twice towards achieving mitigation targets or goals (e.g., by two entities or twice by one entity).

Emission Reduction
A net reduction in anthropogenic emissions by sources.

Grievance Resolution Mechanism
An independent mechanism for addressing complaints, concerns or disputes related to the mitigation activities in a fair, effective and transparent way.

Jurisdictional REDD+ Activities
Mitigation activities undertaken at national or sub-national scale by jurisdictions in developing countries to reduce emissions from deforestation and forest degradation, as well as the sustainable management of forests and the conservation and enhancement of forest carbon stocks.

Mitigation Activity
An activity that reduces anthropogenic emissions of a greenhouse gas or enhances removals by sinks relative to emissions in the activity's baseline scenario and seeks registration and issuance of carbon credits under a carbon-crediting program. The term refers to activities that may be implemented at different scales, including projects, programmatic approaches, policies or jurisdictional REDD+ activities. They may also be implemented at one or more sites.

Mitigation Activity Proponent
The legal entity requesting the registration of a mitigation activity and issuance of carbon credits under a carbon-crediting program. The mitigation activity proponent may be a public or private entity.

Non-Monetary Benefits
Goods, services or other benefits funded from carbon credit revenues, or directly related to the implementation and operation of the mitigation activity, that provide a direct incentive to Partners to help implement the activity and can be monitored in an objective manner (e.g., technical assistance, capacity building, and in-kind inputs or investments such as seedlings, equipment, buildings). These benefits, if any, are included in the Benefit Sharing Plan.

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Non-Permanence
A situation in which mitigation is subject to reversal.

Normative Program Documents
The documents adopted under a carbon-crediting program that specify requirements, procedures, and administrative and operational aspects of the program. These documents include, but are not limited to, general program standards, quantification methodologies, procedures, provisions, manuals, guidance and forms.

Offsetting (Offset)
The compensation for an entity's greenhouse gas emissions within its scope by achieving an equivalent amount of emission reductions or removals outside the boundary or value chain of that entity.

Partners
A subset or group of a mitigation activity's stakeholders (IPLCs involved in or affected by mitigation activity implementation) identified in the activity documentation to receive Monetary and/or Non-Monetary Benefits resulting from the sale of carbon credits or result-based payments of emissions reductions.

Project Scenario
A description of the situation and the outcome that is predicted or assumed to occur in the presence of the mitigation activity to be credited, while holding all other factors constant.

Removal
A net enhancement of anthropogenic removals by sinks.

Retirement
The transfer of a carbon credit to a registry account that permanently removes the carbon credit from circulation. The term retirement applies to the use of the carbon credit by an entity to meet voluntary commitments or compliance obligations. The term is distinct from administrative cancellations.

Validation and Verification Body (VVB)
An independent third-party entity that is accredited for performing validation and/or verification audits. Verification and validation bodies are often also referred to as auditors.

Validation
The process of independent, third-party evaluation of a mitigation activity requesting registration under a carbon-crediting program by an accredited Validation and Verification Body (VVB) against the program provisions of the applicable carboncrediting program.

Verification
The process of periodic independent, third-party ex-post evaluation by a VVB of requests by a registered mitigation activity to issue carbon credits against the program provisions of the applicable carbon-crediting program.

Vintage
The year in which the carbon emission reduction took place. Because the verification

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process can take two to three years from project/program inception, projects/programs may generate credits for already-reduced emissions

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Workshop Overview

On January 31 and February 1, 2024, 60+ experts from across the carbon market ecosystem—policymakers, practitioners, academia, and more—convened at Cambridge University to discuss issues and challenges related to permanence and durability and identify potential solutions for the continued development of the market.

The discussion at the Cambridge Permanence and Durability Workshop will serve as an input to the IC-VCM CIWP on Permanence. After publishing its Core Carbon Principles and corresponding Assessment Framework in July 2023, the IC-VCM launched the CIWPs to consider key issues related to the future of the carbon markets and to develop recommendations to inform the next iteration of IC-VCM CCP label requirements.

The Cambridge Permanence and Durability Workshop was the first step of a rigorous stakeholder engagement process leading up to the release of the revised IC-VCM standards in 2026.

The workshop was designed with two objectives in mind:

- **Develop a shared understanding among the expert community** on 1) approaches to addressing the issue of permanence for carbon crediting purposes, and 2) on key open questions or gaps in understanding or knowledge.
- **Develop recommendations for addressing permanence** (including risk assessment, allocation, and mitigation) that will result in **"investment-grade" carbon credits that maintain scientific rigor in delivering climate impact while balancing complex policy, legal, financial, equity, and implementation tradeoffs.**

These objectives tie directly to the IC-VCM's overarching mission of setting definitive global standards using the best available science and expertise, so that high-quality carbon credits can serve as a financing mechanism for genuine and additional greenhouse gas (GHG) reductions and removals.

In line with these objectives and the IC-VCM mission, the workshop had two distinct parts: 1) expert presentation and discussions related to permanence and durability and 2) a "systems design" exercise. The panel discussions addressed a range of topics related to permanence, including scientific perspectives on the economic and atmospheric impact and value of carbon storage, to the role of the financial sector and insurance in carbon markets, to an overview of how standard-setting bodies currently address permanence and durability. During the systems design exercise, workshop participants worked in small groups to develop recommendations for how permanence and durability should be addressed by various market actors.

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Workshop Discussion Insights

The Cambridge Workshop provided a forum to share perspectives on permanence and durability in carbon markets. Ahead of the workshop, the organizers conducted informal, 30-minute interviews with 30+ attendees to solicit their opinion on the unresolved challenges regarding permanence and durability, promising measures to mitigate reversal risk, and the role of the IC-VCM in issuing permanence and durability guidance that allows for the market to scale. During the workshop, attendees participated in panel discussions on a variety of topics related to permanence and durability and had the opportunity to share their reflections in small-group and large-group formats.

As expected, attendees' views differed depending on their role in the ecosystem and beliefs about the role of carbon credits in driving climate benefits; however, workshop participants noted that there appeared to be more consensus than previous discussions about permanence and durability. Below is an aggregated, anonymized synthesis of views about key topics expressed during pre-workshop interviews and discussion during the Workshop itself.

Shift From Permanence to Durability

In the pre-workshop interviews, participants were quick to point out the importance of semantics when discussing reversal risk, noticing the unique name of the workshop: The Cambridge Permanence *and* Durability Workshop. While "permanence" has historically been the accepted term for this type of risk, several attendees advocated for shifting to a discussion of durability as a better way to conceptualize the issue. This change in language reflects that there are several different factors that determine how long the benefits of a particular mitigation activity can be expected—some of which are easier to control for than others—and that there is an inherent unknown when discussing the length of time that an activity sequesters carbon. As one participant noted in their reflections, "what's important is that carbon stays sequestered for as long as it matters."

For projects deemed to have a material risk of reversal, the IC-VCM currently sets that threshold at 40 years, but in a panel with members of carbon crediting programs, it was evident that each body takes a different approach (see Current Practice Among Carbon Crediting Programs). Throughout this panel discussion and the workshop, the ideal minimum threshold to establish "permanence" was debated, with participants voicing support for anything from 20 years to 100+ years. Notably, there were no participants that advocated for a 1,000-year permanence standard, even though there are market proponents of this approach.[2] Vocal opposition towards longer thresholds came from the project developer community, noting that in many parts of the world (especially the Global South), there is no legal framework in place or appetite among landowners to

[2] Hausfather, Zeke, Freya Chay, Jeremy Freeman, and Danny Cullenward. "The SBTi Net-zero Standard Should Include Guidance on Carbon Removal." *CarbonPlan* (blog). November 22, 2022. https://carbonplan.org/research/sbti-carbon-removal.

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sign a contract with such a long duration. Participants specifically called out the Māori regulations, where any leases longer than 21 years require court approval, and perpetual leases are often structured as 21-year leases with the option to renew at each 21-year term.[3]

While there was ultimately no consensus around the appropriate threshold to define "permanence," and workshop participants recognized that there is no "right answer" to defining permanence, there was recognition for a need to be more transparent about the anticipated duration of a particular mitigation activity. Attendees noted that there could be several factors including environmental, legal, and social risk that served as an input into developing the expected duration of a mitigation activity. Workshop participants also noted that the market currently lacks a mechanism to address reversal risk after a project's expected duration and suggested that various actors—from the IC-VCM, to carbon crediting programs, to national governments—develop a system to monitor and account for longer-term reversal risk.

Issue of Fungibility

In the pre-workshop interviews, nearly every interviewee raised the issue of fungibility (i.e., equating one tonne of carbon from one project to a tonne of carbon from a different project or emissions source with a different duration) as one of the top open questions around permanence and durability. There remains no market-wide consensus on how purchasers or regulatory bodies should address varying levels of reversal risk or duration associated with credits from different activity types.

Some participants argued that once minimum reversal risk mitigation measures were in place (e.g., a buffer pool, insurance, etc.), durability should not matter to carbon credit purchasers, assuming the credit is backed by a credible system that "makes the atmosphere whole" in the event of a rerelease of stored carbon. Their rationale was rooted in the severity of the climate emergency that we are now facing, given slow and insufficient progress against the 1.5C° threshold agreed to under the Paris Agreement.[4] They advocated for guidance that matched the urgency of the crisis, arguing that there is climate benefit that accrues at most durations of carbon sequestration. If society has the privilege of debating permanence in 50 years, they argue, then we have averted the worst of the anticipated climate disasters. These participants often highlighted that it is impossible to guarantee the permanence of any carbon removal or reduction outcome, given that legal and market structures do not exist to ensure longevity beyond several decades, and there is significant uncertainty related to climate or human conditions in hundreds, let alone thousands, of years.

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[3] "Māori Land: What is a Whenua Lease?" Ministry of Māori Development. Accessed March 4, 2024. https://www.tupu.nz/en/kokiri/whenua-leases/what-is-a-whenua-lease.

[4] "New Analysis of National Climate Plans: Insufficient Progress Made, COP28 Must Set Stage for Immediate Action," United Nations Framework Convention on Climate Change, last modified November 14, 2023, https://unfccc.int/news/new-analysis-of-national-climate-plans-insufficient-progress-made-cop28-must-set-stage-for-immediate.

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Others emphasized the importance of ensuring that claims of reduction or removal matched the actual emissions liability and expressed skepticism at the notion of a 1:1 equivalence of all carbon credits. This logic was frequently mentioned in the context of atmospheric equivalence of carbon emissions, noting that carbon credit buyers use their purchase to offset a tonne of carbon dioxide emitted from fossil fuel combustion, which could remain in the atmosphere for up to 1,000 years.[5]

Carbon credit buyers who believe that a credit backed by a short-term activity is an equivalent offset to their carbon dioxide emission could contribute to negative climate impacts through increased cumulative warming if reversal occurs and is not compensated for, or if commensurately more climate damage is done from the emission than from the activity that underpins the credit retired to address that emission. Participants also noted implementation concerns around a market wide approach to fungibility. Given that several countries have already implemented (or are in the process of implementing) regulation for their carbon markets, it would be challenging to retroactively align around one standard.

Some participants noted that differentiation among carbon credits could be a positive. If buyers better understood the expected durability of a particular mitigation activity, it could aid them in developing a balanced portfolio of credits from different activity types. Attendees noted the value of a portfolio approach when discussing the need for investment in more permanent, engineered solutions to carbon capture and storage to scale in time to help achieve climate goals.

Overall, there was consensus that IC-VCM should coordinate closely with other market oversight bodies, policymakers, and other market actors in developing any guidance that could impact the perceived fungibility of carbon credits.

Need for Increased Investment

There was broad recognition that carbon markets needed to shift towards being a more investable asset class to achieve the scale needed to meet climate goals. In their 2021 blueprint, the Task Force for Scaling Voluntary Carbon Markets (the IC-VCM's predecessor organization) highlighted the need for a 15-fold increase in the supply of carbon credits by 2030 and a 100-fold increase by 2050 to reach targets consistent with the 1.5-degree pathway.[6] This cannot be accomplished without a significant increase in the amount of public and private finance currently being invested in the carbon markets.

Issues of permanence and durability are cited as one of several factors that could be holding back further investment. In a panel discussion on creating an investment-grade asset class, there was consensus that clear and accepted guidance around issues of

[5] Buis, Alan. "The Atmosphere: Getting a Handle on Carbon Dioxide." *National Aeronautics and Space Administration* (blog). October 9, 2019. https://climate.nasa.gov/news/2915/the-atmosphere-getting-a-handle-on-carbon-dioxide/.

[6] *Final Report*. Task Force for Scaling Voluntary Carbon Markets, 2021. https://icvcm.org/wp-content/uploads/2022/03/TSVCM_Report.pdf.

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permanence and durability as well as other core elements of supply (and demand) was crucial in order for private capital to be mobilized.

Panelists also discussed other factors that are inhibiting scale, including the lack of intermediaries that could help large financial institutions better understand what they are investing in. They noted that currently, you have to be very close to a particular project and its diligence to arrive at an investment decision, but in other commodities markets, this is not the case. Panelists noted that intermediaries can also play a role in aggregating and communicating risks using standardized criteria, which could provide clarity and lead to securitization.

In terms of immediate options to increase investment while the financial plumbing develops over the course of the next several years, workshop participants highlighted two suggestions: alignment around standards and long-term offtake. In discussing standards, participants noted that the IC-VCM's Core Carbon Principles were a great first step, and once fully implemented, could serve to unite the market around what high-integrity supply looks like. Attendees also mentioned the importance of long-term offtake given the nature of a (typically nature-based solutions) project developer looks like. With little-to-no collateral to lend against, long-term offtake contracts can help to de-risk project finance.

Improved Compensation Mechanisms

In the most recent Assessment Framework, the IC-VCM endorsed buffer pools as the sole compensation mechanism for avoidable reversal risks, as it was the only approach that had gained widespread market acceptance. It also noted that there are a range of additional reversal compensation mechanisms under development and piloting across the market, including insurance, and that those will be further explored in the context of the Continuous Improvement Work Programs. Workshop participants agreed on the need to improve buffer pool design and operation, while also allowing for the use of alternative compensation mechanisms.

Workshop attendees linked three principal issues to buffer pools when discussing their improvement: risk allocation, appropriate capitalization, and conflict of interest. The issue of risk allocation centers on which parties bear the risk of reversal when. Current IC-VCM guidance places this risk largely on the carbon crediting program or project developer for a defined monitoring and compensation period, after which no one bears the risk. Workshop participants suggested ideas for how risk could be redistributed by requiring purchasers to bear some risk and replace credits that face reversal, creating a public entity to take on the risk after the monitoring and compensation period ends, or simply improving current buffer pool management with "sunset clauses" to transfer risk after a certain period of time or when the organization managing the buffer ceases to exist. Standard setting bodies are also exploring whether long-term liability for monitoring and reversal mitigation could shift to them from the developer at the end of a project's active crediting period. Most participants agreed that project developers or carbon crediting programs should continue to bear the costs of reversal risk, with one participant noting that it is unheard of that a buyer should be responsible for the quality of a product in other industries.

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Several attendees noted that insurance and reinsurance could play a role in helping project developers and carbon crediting programs mitigate reversal risk. For example, through the purchase of an insurance policy, project developers could manage their own reversal risk and eliminate the need for pooled buffer reserves at the carbon crediting program level. Alternatively, carbon crediting programs could work with insurers to take on management of buffer pools themselves, leveraging the advanced analytics capabilities that are in-house at many insurance companies. While the workshop conversation on the role of insurance was largely positive, some participants cautioned that insurance is not a panacea and that the industry is still nascent. While some of the larger insurance conglomerates participate in the space, insurance for carbon credits is still largely specialized, and insurance for reversal risk in particular is still developing.

Another issue raised with respect to buffer pools was appropriate capitalization. A recent analysis of the buffer pool associated with California's cap-and-trade program found that wildfires have already depleted 20% of the credits meant to mitigate reversal risk for the next 100 years.[7] To address capitalization concerns, participants suggested several different ideas (many of which are noted in the Systems Design Exercise Outputs) including: encouraging carbon crediting programs to purchase more durable credits to pair with issuances for shorter-duration activity types, requiring the use of customized risk assessments instead of blanket buffer pool contributions for high-risk mitigation activities, or insurance policies for pooled buffer reserves that payout when the buffer drops below a certain credit count.

Workshop attendees also pointed out the inherent conflict of interest that exists for buffer pools: those who manage buffer pools as a risk mitigation mechanism are also responsible for emitting carbon credits. In a competitive marketplace for carbon crediting programs, standards bodies could be incentivized to lower the amount of buffer contributions required to allow project developers to issue more credits and maximize their revenue. Carbon crediting programs to-date have built in safeguards between those who may interface with project developers and those who manage the risk evaluation process; however, given they are affiliated with the same organization, the risk cannot be mitigated entirely.

Scaling Longer-duration Carbon Sequestration and Storage

Cambridge Workshop attendees universally recognized the role that engineered carbon removals and mitigation activities with longer-duration storage will play in helping to achieve climate goals. Several participants expressed alignment with the Oxford Principles. The Principles allow the use of short-duration offsets as a bridge to longer-

[7] Badgley, Grayson, Freya Chay, Oriana S. Chegwidden, Joseph J. Hamman, Jeremy Freeman, and Danny Cullenward. "California's forest carbon offsets buffer pool is severely undercapitalized." *Frontiers in Forests and Global Change* 5 (August 2022). https://www.frontiersin.org/articles/10.3389/ffgc.2022.930426/full.

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term storage but emphasize that “significant investment in in carbon removal with a low risk of reversal begins now to reach the thousand-fold increase needed by 2050.”[8]

Some workshop attendees viewed IC-VCM’s guidance around permanence as an important input to directing an appropriate amount of investment towards longer-duration storage. For example, this could be done indirectly by setting a high threshold for what qualifies as “permanent” or directly like mandating investment in emerging technologies as part of a pooled buffer reserve. While there was no consensus opinion on how the IC-VCM should approach this issue, it was re-iterated by workshop participants as an important challenge to address in the near-term.

[8] Axelsson, Kaya, Audrey Wagner, Injy Johnstone, Myles Allen, Ben Caldecott, Nick Eyre, Sam Fankhauser, et al. *Oxford Principles for Net Zero Aligned Carbon Offsetting (revised 2024)*. University of Oxford, 2024. https://www.smithschool.ox.ac.uk/sites/default/files/2024-02/Oxford-Principles-for-Net-Zero-Aligned-Carbon-Offsetting-revised-2024.pdf.

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Systems Design Exercise Outputs

The systems design exercise provided space for workshop participants to work in small groups of diverse backgrounds to develop a system that would effectively address permanence and durability risk while achieving three objectives: delivering climate change impact, maintaining an appropriate quality bar for high-integrity supply of carbon credits, and encouraging investment at a scale commensurate with the size of the climate crisis. Each group was asked to describe any risk mitigation and or/compensation mechanisms envisioned in their system, consider the benefits and tradeoffs of adopting the approach, outline any considerations for specific project types, and evaluate their system against the Permanence and Durability Parameters. In total, there were eight different groups, each with their own unique perspective; however, several reversal risk mitigation mechanisms appeared across groups.

Common Threads Throughout Outputs

Adjusting buffer pool requirements to focus on more durable solutions. Three groups mentioned changing buffer pool requirements at the carbon crediting program level over time to focus on solutions with less reversal risk. For example, a given standard or registry would be required to have a minimum percentage (e.g., 15%) of its pooled buffer reserve comprised of credits from low-reversal risk credits (e.g., carbon capture with geologic storage) and this percentage would increase over time until 100% of the pooled buffer reserve was made up of these credits. Adopting this guidance would likely require a change in how carbon crediting programs manage their pooled buffer reserve. Currently, the pooled buffer reserve makeup generally mirrors the portfolio of projects certified by a particular standard. This guidance could see carbon crediting programs work with project developers to contribute credits to the pooled buffer reserve that are not necessarily credits from the program in which they are certifying, depending on the threshold of low-reversal risk credits required.

Contributing to a fund for long-duration credits. A couple of the groups recommended the creation of a fund to invest in long-duration credits. The fund could be capitalized in a variety of ways, including through a potential fee levied on the issuance of short-duration credits or through required contributions to the fund of shorter duration credits that could be liquidated in the event of a reversal and proceeds used to procure permanent CDR. These funds could then be invested in a trust or endowment to compensate for reversals after the conclusion of the 40-year monitoring and compensation period, with a focus on supporting the development of engineered solutions that have a longer duration. Participants noted that this idea recognizes the risks associated with shorter-duration projects and could provide a source of capital to build the capacity of longer-duration projects over time.

Setting a new minimum duration requirement. In the current version of its Assessment Framework, the IC-VCM set a minimum 40-year monitoring and compensation period for projects deemed to have high reversal risk. Several groups aligned around a 100-year monitoring and compensation period as a revised minimum duration requirement; however, the risk allocation and responsibilities could change throughout the period. For example, several groups suggested that project developers be responsible for reversal risk for the first 40 years of the 100-year period but that

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carbon crediting programs or independent bodies be responsible for monitoring and compensation during years 40-100. No groups suggested setting a minimum threshold of more than 100 years.

Creating an independent third party to evaluate or monitor ongoing reversal risk. Many groups recognized the gap in current guidance after the expiration of the 40-year monitoring and compensation period for high-reversal risk projects. Participants noted Verra's recently announced effort to develop a long-term monitoring system for reversal risk, but recognized that there could be efficiencies of scale achieved by sharing this data and possibly implementation structures across crediting programs and centralizing the long-term reversal risk monitoring. This independent, third party could also re-evaluate reversal risk over time to ensure that any compensation mechanism (e.g., buffer pool) is adequately funded as risk profile changes.

Requiring the replacement of credits after the expiration of the monitoring and compensation period. Some participants recommended a systematic adoption of temporary crediting as a solution to addressing reversal risk. After a credit's monitoring and compensation period expires, the buyer, standard, or some other body would be required to replace it with another credit. This credit could be more durable in nature with an eye towards long-term carbon storage or a shorter-duration credit—whichever is more feasible.

Details of Individual Outputs

The details of the proposed approach for each group at the Cambridge Workshop are included below. Responses to each question have been lightly edited for grammar; however, they have been left largely intact as "raw outputs" from the exercise. This was done intentionally to allow the reader to draw their own conclusions from each proposed approach. If a question is left blank, the group did not provide an answer to it. The proposals listed in the order in which they were presented. Inclusion of these proposals here does not indicate an endorsement of these proposals by the convenors of the workshop.

Proposal #1

What is a brief description of your approach?

At the registry level, the crediting program will require that buffer pool contributions to mitigate reversal risk are made from qualified non-reversible credits. Over time, the proportion contributed to the buffer pool from long-term durable storage credits increases to 100%.

The buffer pool is the compensation mechanism, and this can be reinsured as needed. If a reversal exceeds what the project contributed to the buffer, the project must make up the difference. The contents of the buffer pool, including the mechanism for increasing the share of removals, will be in the public domain.

How will this approach achieve objectives of scaling high-integrity supply and investment?

This approach is an on-ramp solution creating a built-in demand pull for high durability removals that increases over time, starting now. It also builds confidence in the

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management of temporary storage because the credits with reversal risk are backed by nonreversible tons, including and increasing percentage of high durability removals.

What are the benefits of this approach? What are the tradeoffs?

Buffer pools are managed transparently in a consistent way at the program level, thereby increasing harmonization among standards & programs. It builds on the existing approach in the market for managing reversal risk. It also addresses the lack of demand signals to increase supply for long-term removals.

The economics of this approach will be reflected in the price for the temporary tons, potentially driving up prices. This does not address the need for accurate oversight of reversals.

What are the considerations for different project types (if any)?

Need to define what credit types are eligible, what is the standardized reversal risk for each project type, appropriate ramp for transition to permanent removal.

What are key assumptions or open questions associated with your approach?

We are assuming there is a higher implied climate value for the permanent tons. Assumes that we have a standardized risk model, appropriate MRV, any other registry issues (i.e., validation and verification body credibility) are solved. Assumes that supply will materialize in response to the demand signal.

Proposal #2

What is a brief description of your approach?

Replace buffer pool with tax on projects according to risk (forward and backward-looking risk assessment updated every five years, but with standard contribution to start). Tax goes into long-term trust/endowment (Nature Bridge Fund?) managed by designee of standard (possibly ONE fund for all standards). Standard takes on durability liability and long-term monitoring years 40-100 (or maybe until all obligations are compensated for), buys credits / makes investments at any time to address reversals (ideally tech removals, though at the beginning maybe a blend). Actual durability reported ex-post, system designed to achieve effective permanence. Ideally every temporary credit gets replaced by a permanent credit (including non-reversals). Accommodates shorter durations with higher upfront contribution.

Other Considerations:

- Endorse 40-year threshold. (Innovation encouraged at standard and developer level to possibly shorten and align with 6.4) developers bears liability for monitoring and reversals, but maybe can also compensate financially as opposed to with credits?
- Exploration of pairing with PACT or vertical stacking (tonne year) or other tactics to manage reversal risk and enable participation during X year period (may require changes at standard level, and needs to be very carefully explored given legitimacy concerns)

How will this approach achieve objectives of scaling high-integrity supply and investment?

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What are the benefits of this approach? What are the tradeoffs?

This answers the question of equivalency, makes a claim credible, opens up funds. Joins nature-based and engineered removals, creating demand for long-term removals. Mirrors what the global science says has to happen (NBS as bridge, tech removals long-term)

What are the considerations for different project types (if any)?

Maybe differential rates/applications for different forest types.

What are key assumptions or open questions associated with your approach?

- This idea needs a lot of modelling and stress testing to see if its feasible
- Engineered solutions will come down in price quick enough and exist at a big enough scale
- Buyers are willing to pay more to accommodate this kind of tax approach
- 40 years is short enough the enable widespread adoption

Proposal #3

What is a brief description of your approach?

- 100-year climate benefit per ton;
- Stewardship model (ownership, liability, beneficiary); accreditation of liable party (e.g. require insurance)
- Reversal managed via buffer pool or, if buffer pool used, re-purchase of ton of at least same vintage year of reversal;
- Data shared for external auditing of climate impact to enable methodology updates

How will this approach achieve objectives of scaling high-integrity supply and investment?

- Consistency of 100 years
- Liability tracking
- Scientific auditing

What are the benefits of this approach? What are the tradeoffs?

- Aligning around 100 years

What are the considerations for different project types (if any)?

- Buffer pool sizing
- Economic efficiency

What are key assumptions or open questions associated with your approach?

- Data that should be shared for auditing
- Liability accreditation

Proposal #4

What is a brief description of your approach?

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Stage 1: Require all standards to standardize risk assessment that determines contribution to buffer pool—e.g., same approach to assessing political, natural risk, management, financial risk, market risks using same data and default value

Stage 2: ICVCM requires over time a transformation of buffer to credits with no reversal risk such as mineralization, etc. When reversal happens, replace with these kinds of credits with - ICVCM requirement to achieve 5%/year shift of buffer pool

Stage 3: Universal buffer hosted by entity such as World Bank, with one approach to when buffer is accessed (currently difference in this among standards, some release over time, some don't). If credits not used within specified period, sold to buyers as a spot product of projects or retired. Sold as product to buyers.

- Size of buffer is conservative but efficient
- Intentional/avoidable risks managed how:
    - Projects with Reversal risk pay a price at issuance based on their level of reversal risk
    - Bonus every crediting period for non-reversals—you get credits out of the buffer—eventually you can sell them. 20-year project with 40-year requirement means higher contribution up front, then payback over time with no reversals
- Reversal detection system managed by an independent body

How will this approach achieve objectives of scaling high-integrity supply and investment?

Standardize management of one of the key risks facing markets, provide incentives for purchase/investment in low-risk reversal projects through buffer

What are the benefits of this approach? What are the tradeoffs?

Feasibility—big change that has to happen in stages over time

What are the considerations for different project types (if any)?

Recognizes benefits of low-reversal-risk projects while not punitive to projects that face limitations in commitment to long-term

What are key assumptions or open questions associated with your approach?

How to refine approach to incentivize avoiding avoidable reversals

Proposal #5

What is a brief description of your approach?

- We are going to assume all credits are high-quality
- Post-closure monitoring will be for 40 years or until functionally stable; if not, then all credits are cancelled
- We set up two 3rd party independent organizations both funded by transaction fees:
    - "CDM methodology panel" which defines the % haircut for each project type
    - The entity that houses the buffer pool

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- All methodologies have conservative discounting for expected losses
- The meth panel will consider issues of social durability
- There is a variable haircut in the form of credits depending on the risk of reversal of the project type
- The haircut goes into the market wide mixed buffer pool
- Periodic haircut updates at least every 5 years based on data informed reversal risk evaluations (i.e. lower risk lower haircut)
- Automatic renewal of the buffer pool, credits are cancelled once the project is no longer issuing credits and monitored for performance.
- Fungibility enables scaling?? and the buffer enables insurance for the risk of reversal.
- Where the credits are cancelled (e.g. due to ceased monitoring) they are also cancelled in the buffer

How will this approach achieve objectives of scaling high-integrity supply and investment?

Risks are diversified; someone can insure and take on the liability; this increases trust confidence and therefore scale. This aligns incentives in terms of who manages the buffer pools because of independence; incentives the project developers to reduce their haircut once they reduce their risks

What are the benefits of this approach? What are the tradeoffs?

It's not perfect, it might not incentivize all removals.

What are the considerations for different project types (if any)?

Big haircuts from the riskier projects, small haircuts from less risky projects.

What are key assumptions or open questions associated with your approach?

We are ignoring issues of backward litigation which could affect the quality of the buffer pool. Continuous renewal of the buffer pool

Proposal #6

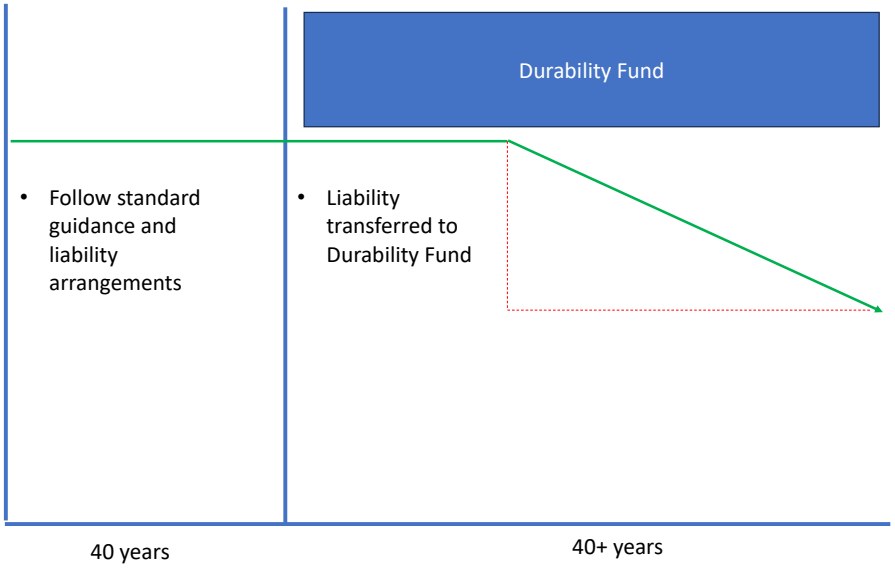
What is a brief description of your approach?

- Design principles
    - Avoid net atmospheric harm but don't paralyze investment
    - Channel funds to durable carbon removal because it is the natural resting place for neutralization claims. Organization portfolios
    - Allow 'cover' to invest in low-durability credits now, because it is critical path to Paris-aligned goals
    - Incentivize nature-based credits that are 'designed' for higher durability
    - Shift long-tail liability for reversals after monitoring period
- Description of approach: long-term permanence fund
    - Durability-weighted portfolio, with a floor for high-durability CDR, growing over time
    - Registry-controlled buffer pool, as in ICVCM proposal, covers unintentional reversals during the at-least 40-year monitoring period

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        - Project developer compensates for intentional reversals
        - Buffer contribution should be highly sensitive to project-specific risk
    - Durability Fund, run by a public body, collects funds (or credits) from every issuance of short-durability credits
        - Contribution is a function of the durability (and thus liability) selected by the project (at least 40 years, up to 100 years)
        - Contribution could be in cash (for re-investment in durable CDR) or in short-durability credits (if we think this avoids price risk in the future)
    - Funds are re-invested in (advanced market commitment for) high-durability credits that are retired

Durability Fund Concept



The Durability Fund (DF) would be managed by a public bodies and could be structured in 2 ways:

Option 1: DF would estimate the potential shortfall for each project and a corresponding number of any credit issuance would be be transferred to the DF ex ante to build up a global buffer pool

Option 2: DF would estimate a financial contribution to buy durable emission reductions to cover estimated shortfalls

How will this approach achieve objectives of scaling high-integrity supply and investment?

What are the benefits of this approach? What are the tradeoffs?

- Benefits
    - Shorter duration credits achieve effective/synthetic permanence if the project contributes to a fund that channels money to advanced commitment high durability CDR.
    - Designing for durability, and taking on longer monitoring (liability) periods are incentivized
    - Liability is limited to the monitoring period
- Drawbacks
    - We don't have a great way of estimating the Durability Fund contribution (whether in cash or credits) today
    - Who should manage the durability fund?

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    - Subjective choice that funneling money towards durable CDR is better than buying multiple lower-durability credits today
    - Potential for remaining atmospheric liability if we don't get the Durability Fund right
- Tradeoffs
    - Subjective choice that funneling money towards durable CDR is better than buying multiple lower-durability credits today
    - Potential for remaining atmospheric liability if we don't get the Durability Fund right

What are the considerations for different project types (if any)?

What are key assumptions or open questions associated with your approach?

Proposal #7

What is a brief description of your approach?

Term limited credits: enabling term contracts, to reduce reversal concerns, infuse market flexibility for scale, and allow a like-for-like comparison of different solutions

Key features of this approach:

- There is flexibility in contract length to allow different project types (e.g. 5-1,000 years) and durability is tied to contract length.
- Each project is labelled by durability, and could be distinguished by other key features (e.g. co-benefits)—giving the market more options and flexibility to set price.
- Horizontal stacking: a company that emits a ton of CO2 buys a horizontal series e.g., can reup after a forestry contract ends to continue investing in that forest or purchase CDR credits in future years if prices come.
- If reversals, project developer must make whole. The buffer pool can be smaller, limited to delivery risk. Managed by standard.

How will this approach achieve objectives of scaling high-integrity supply and investment?

- Accelerates bringing supply online
- Enhances transparency and honesty: does not force projects to claim spurious commitments to "durability" that are predicated on systems that do not yet exist (e.g. monitoring periods far beyond project timeline, etc).
- Gives the market more options and flexibility.

What are the benefits of this approach? What are the tradeoffs?

- Benefits
    - Feasible: works within existing market, legal etc frameworks (e.g. limitations in local legislation regarding contract lengths).
    - Inclusive: allows more smallholders and communities to participate and reduces risks to them.
- Tradeoffs

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    - Requires commitment by corporates/emitters to shoulder the liability and continue horizontal stacking (or see below about transferring liability).
    - Makes it harder for the market to claim carbon neutrality (but this is increasingly not allowed)
    - Possible that it causes crashing in prices for shorter-duration contract terms—but this needs market research

What are the considerations for different project types (if any)?

What are key assumptions or open questions associated with your approach?

- An entity (e.g. Global Carbon Market Utility) would likely be needed to manage the liabilities created by buyers for requirement to repeatedly purchase credits for the same emission.
- This doesn't "solve" permanence, but instead stops pretending that the current system is permanent. We are acknowledging reality.

Proposal #8

What is a brief description of your approach?

- The approach is a 40-year rule from issuance date with some caveats around contracting not being the only consideration for durability. Every tonne has a 40-year durability according to this approach. The consideration here is that commercial plantation harvest might not be able to meet those standards because of replanting laws, so this is a solution to part of the problem and exceptions might be in order.
- Stress testing of Buffer pools
- The standards have a self-monitoring mechanism every 5 years for continuous monitoring of losses that could result in reversals.
- Transition and longevity plan

How will this approach achieve objectives of scaling high-integrity supply and investment?

This approach is not a big departure from the ICVCM position. We believe that this as a bare minimum requirement for standards to follow. Also the added years of longevity could be an answer to the integrity question.

What are the benefits of this approach? What are the tradeoffs?

Tradeoffs are of the science. It is very clear that the actual claim should be time stamped, but market acceptability to this complexity is uncertain.

Benefits are project level flexibility. Very important for market access

What are the considerations for different project types (if any)?

Considerations if any should be the way this durability period is established. Contracts should not be the only basis for this period. Could be considered differently for developed and developing economies. Harvest ARR should be looked at differently.

What are key assumptions or open questions associated with your approach?

Should high durable credits be labeled? Is 40 years after crediting the answer?

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